

MLP vs. Traditional Baselines for Spray Penetration

Mie Spray Postprocessing Project

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Scope of This Overview

Role

This short deck is the bridge between method families. The detailed evidence is split into three thesis decks: baseline comparison, engineered-feature scaling, and probabilistic calibration.

Current headline

On the common external clean diagnostic set (795 561 points, 33 845 trajectories), the $\Delta P^{0.5}$ Stage-3 MLP five-seed mean beats calibrated Hiroyasu–Arai, Naber–Siebers delay, and sparse heteroscedastic GP on RMSE, MAE, and P95 absolute error.

Summary of Recent Experiments

Recent Progress & Evaluations:

- **Traditional Baseline Comparison:** Benchmarked MLP against empirical formula curve fitting (Naber-Siebers, Hiroyasu-Arai).
- **Stability Testing:** Conducted 5-seed Stage-3 evaluation to separate a real effect from seed luck.
- **Machine Learning Baseline:** Implemented sparse heteroscedastic Gaussian Process regression as a non-parametric uncertainty baseline.
- **Feature Engineering Upgrade:** Replaced sparse raw pressure/diameter axes with $A = \Delta P^{0.5} \rho_a^{-0.25} d_n^{0.5}$.
- **OOD Testing:** Added leave-one-nozzle-out evaluation to test unseen nozzle-family generalization.
- **Probabilistic Calibration:** Evaluated reliability, ECE, CRPS, PIT, and Gaussian coverage.

Metric	Meaning
RMSE	Penalizes large point-prediction errors; primary accuracy metric.
MAE	Typical absolute millimetre error; less tail-sensitive than RMSE.
P95 abs. err.	95th percentile of $ \hat{y} - y $; reports tail risk.
Bias	Mean signed error; detects systematic over-/under-prediction.
1σ , 2σ coverage	Fraction with $ y - \mu \leq k\sigma$; calibrated Gaussian targets are 0.683 and 0.954.
ECE	Mean reliability-diagram gap over lower-tail probabilities; lower is better.
CRPS	Proper probabilistic score in mm; lower means better calibration-sharpness trade-off.

Traditional Baseline I: Naber–Siebers Correlation

Formulation: The Naber–Siebers model correlates spray penetration $S(t)$ with physical parameters:

$$S(t) = K \cdot \underbrace{\left(\frac{\Delta P}{\rho_a} \right)^{0.25}}_{\text{Scaling Factor } A} \sqrt{d_n} \cdot \sqrt{\max(t - t_d, 0)}$$

Fitted Parameters:

- K : Global scaling constant.
- t_d : Fitted temporal delay (accounting for hydraulic lag).

Image Processing Data Utilized:

- **CDF Penetration Series:** Cleaned, delay-aligned penetration trajectories from top-view Mie scattering.
- **Cone Angle θ (Optional Variant):** Incorporates spreading factor $\frac{1}{\sqrt{\max(\tan(\theta/2), 10^{-9})}}$ extracted directly from video segmentation.

Traditional Baseline II: Hiroyasu–Arai (1990) & Zhou (2019)

Formulation (Multi-stage):

- **Early Stage** ($t < t_b$): $S = k_v \sqrt{\frac{2\Delta P}{\rho_l}} t$
- **Late Stage** ($t \geq t_b$): $S = k_p \left(\frac{\Delta P}{\rho_a} \right)^{0.25} \sqrt{d_n t}$
- **Breakup Time**: $t_b = k_{bt} \frac{\rho_l d_n}{\sqrt{\rho_a \Delta P}}$
- **Zhou Extension (Post-injection, $t > 2t_i$):**

$$S = \sqrt{2} k_p \left(\frac{\Delta P}{\rho_a} \right)^{0.25} \sqrt{d_n t_i^{0.25}} (t - t_i)^{0.25}$$

Fitted Parameters:

- k_v, k_p, k_{bt} calibrated on the dataset (or literature defaults).

Machine-Learning Baseline: Sparse Heteroscedastic GP

Model role

The GP is a strong non-parametric baseline for smooth functions and uncertainty. It tests whether the MLP only wins because no ML baseline was attempted.

$$f(x) \sim \mathcal{GP}(m(x), k_{\theta}(x, x')), \quad \hat{S} = S/A_{\text{scale}}.$$

- Sparse variational GP avoids exact GP $O(n^3)$ scaling.
- Mean GP predicts penetration; a residual/log-variance stage supplies heteroscedastic $\sigma(x, t)$.
- In the clean benchmark, GP is closer to nominal coverage, but MLP is more accurate.

External Clean Headline Numbers

Model	RMSE	MAE	P95	1σ	2σ
Hiroyasu–Arai calibrated	9.974	7.686	20.493	0.536	0.886
Naber–Siebers delay	8.840	6.780	17.922	0.620	0.893
Sparse heteroscedastic GP, 5-seed mean	6.898	5.206	14.365	0.670	0.950
Stage-3 MLP $\Delta P^{0.5}$, 5-seed mean	6.211	4.587	13.029	0.774	0.981

Relative RMSE reductions for MLP: 37.7% vs HA, 29.7% vs NS, and 10.0% vs GP. The MLP is conservative on clean-set coverage; GP is closer to nominal coverage.

Source: Thesis/generated/baseline_comparison_20260509/headline_comparison.csv

Why Compare against these Baselines?

- **Rigidity of Traditional Models:** Empirical formulas lock the penetration curve into $\sqrt{t}$ or $t^{0.25}$ dependencies, ignoring specific internal nozzle geometry or complex post-injection pressure wave dynamics.
- **Advantage of MLP:** With the **upgraded engineered features** (e.g., $A = \Delta P^{0.5} \rho_a^{-0.25} d_n^{0.5}$), the MLP is anchored to physical dimensions but retains the capacity to learn non-linear residuals.
- **Data Utilization:** Naber-Siebers optional variant relies on the explicit video-measured cone angle θ . The MLP aims to predict penetration mapping directly from controllable condition parameters, without needing "oracle" visual features during inference.